Quantification: 535_z13c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	15
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2489.6	1209	2573.59	px^2
major_axis_length_px	lacuna shape	77.0152	70.3083	46.2331	px
minor_axis_length_px	lacuna shape	37.1679	32.8395	20.0419	px
aspect_ratio	lacuna shape	2.0405	2.0691	0.4967	unitless
eccentricity	lacuna shape	0.8329	0.8754	0.1151	unitless
solidity	lacuna shape	0.8166	0.8008	0.0821	unitless
roots_count	headline	7.2	6	4.7087	count
ring_length_r30_px	headline	243	225	140.37	px
ring_length_r60_px	reported	497.4	420	248.219	px
owned_length_px	ownership-dependent	549.429	403.948	509.412	px
edge_count	ownership-dependent	26.3333	20	23.3901	count
mean_edge_length_px	ownership-dependent	21.5498	20.7424	10.0993	px
perimeter_px	normalised	240.806	194.894	175.051	px
ring_area_r30_px2	normalised	8360	6125	4050.14	px^2
ring_area_r60_px2	normalised	18519.3	17204	6811.93	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9969	1	0.0092	unitless
ring_density_r30	normalised	0.0286977	0.0301183	0.00839001	px^-1
ring_density_r60	normalised	0.0268021	0.0269705	0.00857598	px^-1
roots_per_100px_perimeter	normalised	3.3156	2.886	1.8861	per 100 px
ring_attached_length_r30_px	network v2	213.8	189	141.543	px
ring_attached_length_r60_px	network v2	363.2	349	263.853	px
ring_length_w_r30_px	network v2	274.812	253.066	160.625	px
ring_length_w_r60_px	network v2	562.878	481.517	283.621	px
sholl_crossings_r10	network v2	6.1333	6	4.0685	count
sholl_crossings_r20	network v2	6.2667	6	3.5146	count
sholl_crossings_r30	network v2	7.2	6	3.821	count
loop_count	loops, report only	1.1333	1	1.1872	count
loop_edge_share	loops, report only	0.1058	0.125	0.0976	share
ring_width_mean_r30_px	width	7.4221	7.669	1.1568	px

Per field
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measure	value	units
analysed_area_px2	1.01123e+06	px^2
total_skeleton_length_px	21279	px
canalicular_length_density_per_px	0.0210426	px^-1
junction_count	371	count
junction_density_per_px2	0.00036688	px^-2
lacuna_count	15	count

lacunae_per_px2	1.483e-05	px^-2
skeleton_component_count	351	count
mean_component_length_px	60.6239	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0236472	px^-1
field_density_without_flagged_per_px	0.0205527	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	798	px
unattached_component_count	49	count
canalicular_length_density_attached_per_px	0.0210426	px^-1
field_length_density_with_unattached_per_px	0.0218318	px^-1
loop_count	28	count
loop_edge_share	0.087	share
edge_count	1080	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.2575	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19723	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	509	42.4483	19.3806	2.1902	0.8897	0.7313	1.0605	62.053	174.511	False
2	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
3	424	32.3309	18.5247	1.7453	0.8196	0.7837	-0.9381	141.42	858.868	False
4	424	39.0325	14.1936	2.75	0.9315	0.9002	-0.6027	429.66	215.08	False
5	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
6	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
7	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
8	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
9	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
10	9054	174.453	85.8111	2.033	0.8707	0.6586	0.1605	649.475	679.118	False
11	1209	70.3083	25.9548	2.7089	0.9294	0.8103	-1	737.055	638.021	False
12	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
13	543	31.4075	23.6295	1.3292	0.6588	0.9282	0.087	780.65	298.357	False
14	455	27.4749	22.5328	1.2193	0.5722	0.9362	1.1384	784.457	318.629	False
15	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	186	464	511.245	24	21.3019	7.0114
2	9	393	868	1203.06	58	20.7424	7.1937
3	2	249	605	229.853	17	13.5208	6.7267
4	6	155	389	597.186	32	18.6621	9.0159

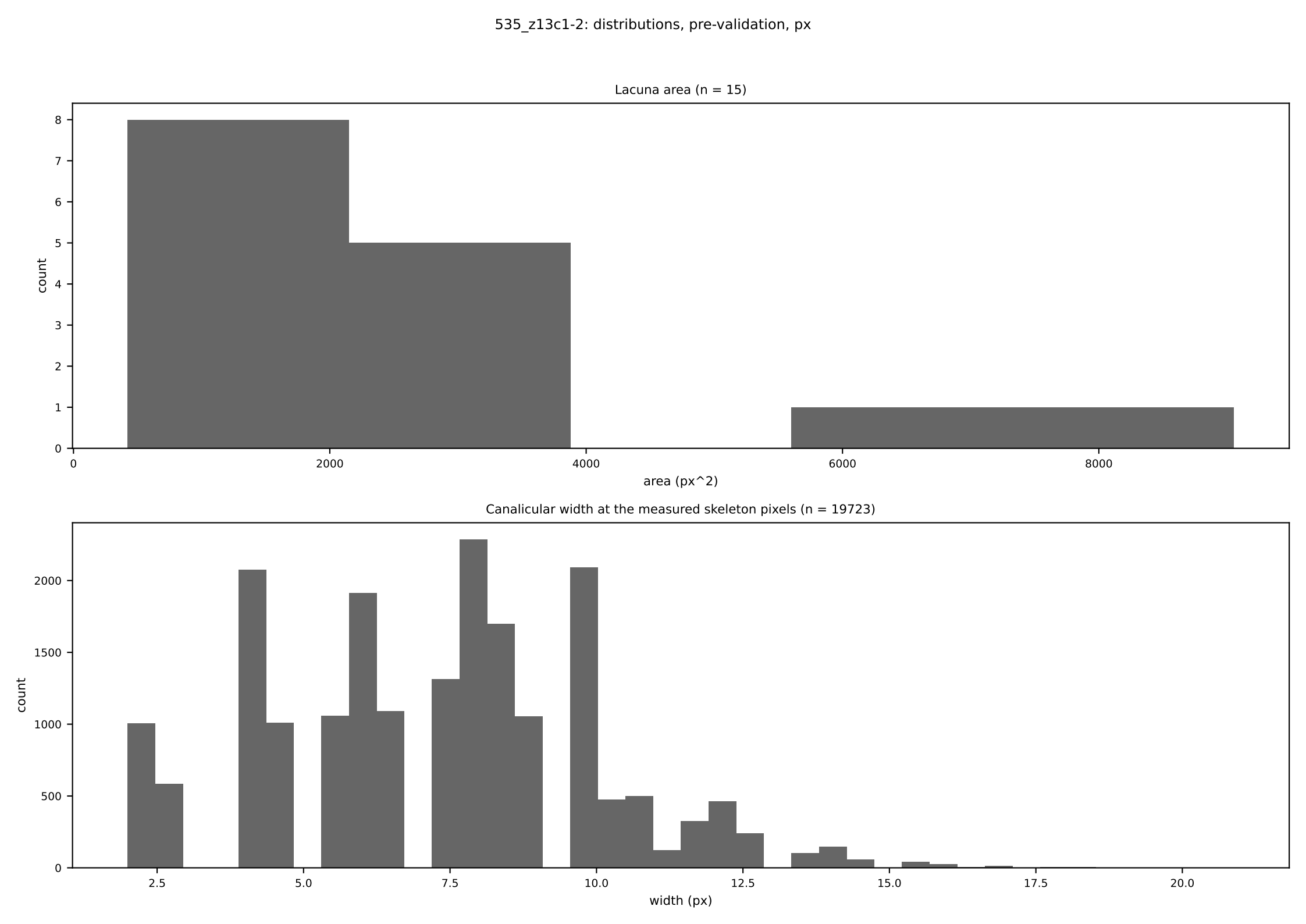
5	11	225	420	769.826	33	23.328	7.6
6	16	453	812	867.252	49	17.699	8.3931
7	7	261	416	225.024	12	18.752	6.6633
8	3	204	417	202.64	7	28.9485	8.5039
9	12	263	531	499.031	24	20.793	7.9127
10	12	519	1027	1994.81	88	22.6682	7.8898
11	5	76	122	211.995	4	52.9987	6.3016
12	13	390	559	403.948	18	22.4416	7.669
13	2	93	312	127.669	8	15.9586	8.3544
14	3	84	193	392.078	20	19.6039	7.829
15	1	94	326	5.8284	1	5.8284	4.2665

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	125.882	6125	17204	1	0.9656	0.0303674	0.0269705	4.7664
2	311.848	10999	24930	1	0.9885	0.0357305	0.0348175	2.886
3	101.569	5469	14333	1	1	0.0455294	0.0422103	1.9691
4	94.5685	5399	12868	1	1	0.028709	0.03023	6.3446
5	145.782	5773	13561	1	1	0.0389745	0.0309712	7.5455
6	433.103	13611	21359	1	1	0.0332819	0.0380168	3.6943
7	326.563	11164	23645	1	1	0.0233787	0.0175936	2.1435
8	270.936	10229	21506	1	1	0.0199433	0.0193899	1.1073
9	264.007	10205	23680	1	1	0.0257717	0.022424	4.5453
10	731.429	16860	33118	1	1	0.0307829	0.0310103	1.6406
11	194.894	5462	13139	1	1	0.0139143	0.00928533	2.5655
12	328.877	11820	23768	1	1	0.0329949	0.023519	3.9528
13	96.0833	3803	10045	1	1	0.0244544	0.0310602	2.0815
14	85.734	2789	7600	1	1	0.0301183	0.0253947	3.4992
15	100.811	5692	17034	1	1	0.0165144	0.0191382	0.992

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	157	275	213.51	530.659	3	4	7
2	385	726	442.09	995.377	10	9	12
3	245	349	279.066	672.487	8	5	7
4	128	369	181.681	451.961	3	5	6
5	215	378	253.066	481.517	6	6	6
6	419	763	519.517	925.423	13	9	13
7	189	210	288.309	456.777	6	12	6
8	186	318	226.095	460.948	7	8	5
9	249	430	303.078	599.286	7	8	8
10	500	935	587.73	1165.08	14	12	14
11	69	95	86.8406	142.368	1	2	1
12	316	361	447.475	633.428	8	8	11
13	70	131	103.012	357.492	2	1	6
14	47	76	89.1127	208.125	1	3	2
15	32	32	101.598	362.25	3	2	4


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 15)
count
area (px^2)
0
1
2
3
4
5
6
7
8
0
2000
4000
6000
8000
Canalicular width at the measured skeleton pixels (n = 19723)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0